2/11/2026.

→ while loop inside while loop.

Syntax :

Initialization of outer while :

while condition :

    Initialization of inner while

    while condition (inner) :

        Statements of inner while

    Statements of outer while.

Ex :
```
i=1
while (i<=5):
    j=1
    while (j<=5):
        print("*", end="")
        j=j+1
    print()
    i+=1
```

3/11/2026.

→ for loop inside while loop.

Syntax :

Initialization

while condition :

    for ptr_var in range(str, spr, ss):

        Inner loop statement

    Statements of while

    Inc/dec of while.

Ex :
```
i=1
while i<=5:
    for j in range(1,6):
        if i==1 or i==5:
            print("*", end="")
        elif j==1 or j==5:
            print("*", end="")
        else:
            print(" ", end="")
    print()
    i=i+1
```

→ while loop inside loop for

Syntax :

```
for ett-var in range (start , stop):
    initialization of while
    while condition:
        statements of while
        increment / decrement of while
    statements of for
```

Ex :

```
for i in rang(1, 10):
    j=1
    while j<=9:
        if j==5-(i-1) and i<=5:
            print ("*", end="")
        elif j==5+(i-1) and i<=5:
            print ("*", end="")
        elif j==i-4 and i>5:
            print ("*", end="")
        elif j==14-i and i>5:
            print ("*", end="")
        else:
            print (" ", end=" ")
        j=j+1
    print()
```